UCT41042 – Practical for Introduction to Smart System

Department of Information & Communication Technology

Faculty of Technology

South Eastern University of Sri Lanka

Lab Sheet No: 12

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Title: Smart Street Light Control and Illumination Monitoring System using ESP32, LDR & Supabase.

Aims:

To design and implement an intelligent street light automation system using an ESP32 microcontroller and LDR sensor. The system automatically controls the light based on ambient conditions, displays status on an I2C LCD, sends real-time data to Supabase, and includes a mobile application to remotely monitor light intensity and street light status.

Task 01: Smart Street Light Monitoring System Setup

1. Project Setup in Wokwi / VS Code
    a. Name the project **SmartStreetLight_ESP32_Supabase**.
2. Circuit Design
    a. Sensor Connection
        i. Connect LDR (Light Dependent Resistor) to ESP32 analog pin.
    b. Output Devices Connection
        i. Connect RGB LED to ESP32 GPIO pins.
        ii. Connect a simple LED (simulating street light) to GPIO pin.
        iii. Connect I2C LCD (16x2) to ESP32.
3. Library Integration
    a. LiquidCrystal_I2C (External)
    b. ArduinoJson (External)
    c. WiFi (Built-in)
    d. HTTPClient(Built-in)

4. Supabase Configuration
    a. Create a new Supabase Project.
    b. Database Setup.
    c. Enable API Access.
5. Programming Logic Development in Wokwi / VS Code
    a. Initialize Wi-Fi connection.
    b. Initialize I2C LCD.
    c. Read analog value from LDR and convert it to light intensity percentage
    d. Display light intensity (%) and street light status on the LCD.
    e. RGB LED Logic (Light Condition Indication)
        i. Define thresholds:
            1. Bright (Day) → Intensity > 70%
                a. LED behavior: Green ON (Street light OFF)
            2. Moderate (Twilight) → 30% – 70%
                a. LED behavior: Blue ON
            3. Dark (Night) → < 30%
                a. LED behavior: Red ON (Street light ON)
            4. Very Dark → < 15%
                a. LED behavior: Red BLINKING
    f. Automatically turn ON/OFF the street light LED based on light level.
    g. Send light intensity, status, and LED state to Supabase every 5 seconds.
6. Simulation & Testing in Wokwi / VS Code

Task 02: Mobile Application Development for Real-Time Monitoring

1. Project Setup in Android Studio
    a. Create a new project named **SmartStreetLight_App**.
    b. Choose Flutter or Native Android (Java/Kotlin).
2. Supabase Integration in Mobile App
    a. Connect the app to the same Supabase project used in Task 01.
    b. Add dependency:
        i. Flutter: supabase_flutter
        ii. Java/Kotlin: Supabase Rest API via Retrofit / HTTP client
3. App UI Design
    a. Create a clean dashboard screen with:
        i. Current Light Intensity (%) displayed prominently.
        ii. Street Light Status (ON/OFF).
        iii. RGB-like visual indicator or colored card for current condition.
        iv. Timestamp of last update.
    b. Add Real-time auto-refresh or manual refresh button.
4. Programming Logic Development
    a. Initialize Supabase Client.
    b. Fetch Real-Time Data
    c. Update UI Logic.
    d. Error Handling
5. Testing the Mobile Application